Report of 2026 Research Activities

State of Alaska Fish Resource Permit CF-25-014

NOAA Fisheries Alaska Fisheries Science Center – RACE Division

Activity Title:

Eastern Bering Sea Continental Shelf Bottom Trawl Survey of Groundfish and Invertebrate Resources

Introduction:

The Resource Assessment and Conservation Engineering (RACE) Division of the National Oceanic and Atmospheric Administration's (NOAA) Alaska Fisheries Science Center (AFSC) conducted the Eastern Bering Sea Continental Shelf Bottom Trawl Survey of Groundfish and Invertebrate Resources from 31 May to 13 August, 2026. This cruise continued the annual series of eastern Bering Sea (EBS) crab and groundfish stock assessment surveys, which began in 1971. 'The standard survey covered the Bering Sea shelf between the depths of 20 and 200 m from Bristol Bay to St. Matthew Island. The northern Bering Sea survey, which usually covers the northwestern Bering Sea shelf from Norton Sound to the Bering Strait, is conducted biennially and next scheduled in 2027. The standard survey covered the Bering Sea shelf between the depths of 20 and 200 m from Bristol Bay to St. Matthew Island. The northern Bering Sea survey, which usually covers the northwestern Bering Sea shelf from Norton Sound to the Bering Strait, is conducted biennially and next scheduled in 2027..

The primary objectives of this survey were to provide the following: 1) data on the distribution, abundance, and biological condition of commercially important groundfish and crab species for the North Pacific Fishery Management Council, 2) catch per unit effort (CPUE) and size composition data for the commercial fisheries of the U.S., and 3) support for ongoing studies on the biology, behavior, and dynamics of key ecosystem components.

Methods:

For the standard survey, sampling at survey stations was coordinated between two chartered commercial fishing vessels, the FV *Northwest Explorer* and FV *Alaska Knight*. Both vessels are house-forward trawlers with stern ramps. Stations were located 20 nm apart and two standard survey stations were located within Alaska state waters (F-20 and H-16). The 26 stations in high-density sampling areas previously located near the Pribilof Islands and south of St. Matthew Island were not included in the standard list of survey stations, as a recent analysis by DeFilippo

et al. (2023) concluded that removal of these stations would have only minor impacts on the precision and accuracy of groundfish biomass estimates. Thus, in the interest of maximizing the efficient use of survey resources, sampling was discontinued in the high-density sampling zones around St. Matthew Island and the Pribilof Islands. The current EBS survey design therefore consisted of 350 planned stations, as opposed to the 376 planned stations sampled annually prior to 2024.

Bottom trawls used for sampling were 83-112 eastern trawls. These nets have a 25.3 m (83 ft) headrope and a 34.1 m (112 ft) footrope. They were towed behind 816 kg, 1.8 X 2.7 m, steel "V"-doors and paired 54.9 m (180.1 ft) dandylines. Each lower dandyline had a 61 cm chain extension connected to the lower wing edge to improve bottom-tending characteristics.

A digital bathythermograph was attached to the headrope and deployed with each trawl, resulting in oblique depth/temperature profiles of the water column. A bottom contact sensor provided data to assess bottom-tending performance of the net and to determine when the footrope was in contact with the seafloor. Net mensuration sensors were used to assess trawl performance and to provide net geometry data to calculate the area swept by the trawl.

For additional information on survey methods and protocols, see Markowitz *et al*. (2025).

Results:

The FV *Northwest Explorer* and FV *Alaska Knight* conducted 355 bottom trawls during the standard EBS survey.. Of those trawls, 5 were determined to have unsatisfactory performance, resulting in retowing at those stations for acceptable performance. A further 92 tows were conducted to support special projects including survey modernization efforts and testing upgraded Marport sensors.

During the standard EBS survey, the scientific crew recorded 127,861 randomly selected length measurements by sex from priority fish species. Additionally, crab measurements (lengths/widths) were collected in the EBS . Shell conditions were assessed as well. Sagittal otoliths were extracted from 13,405 fish from 15 targeted species in the EBS. Correlated length/weight measurements were recorded during otolith collection. A total of 3,669 fish stomachs from the EBS were extracted and preserved for food habits analysis.

Whole specimens of selected fish and invertebrate species were preserved for use in identification training programs and other research. Various tissue samples were removed and preserved for approved research projects. A total of 60 fish and invertebrate taxa were identified in catches within the 3-mile territorial waters (Table 1), and 0 voucher specimens were collected (Table 2).

Table 1: EBS Catch Summary Table:
Within 3-mile Territorial Waters

Common Name	Scientific Name	Weight (kg)	Number
northern rock sole	*Lepidopsetta polyxystra*	1720.392	5160
yellowfin sole	*Limanda aspera*	488.081	1885
walleye pollock	*Gadus chalcogrammus*	434.702	364
yellow Irish lord	*Hemilepidotus jordani*	363.861	472
Pacific cod	*Gadus macrocephalus*	272.12	83
purple-orange sea star	*Asterias amurensis*	266.809	4041
Aleutian hermit	*Pagurus aleuticus*	234.221	2829
sea football	*Cucumaria fallax*	217.371	251
hairy hermit crab	*Pagurus capillatus*	196.127	2357
arrowtooth flounder	*Atheresthes stomias*	167.217	506
Oregon triton	*Fusitriton oregonensis*	162.654	1697
empty gastropod shells	*N/A*	155.11	N/A
Pribilof whelk	*Neptunea pribiloffensis*	115.036	943
N/A	*Metridium sp.*	93.349	471
purple hermit	*Elassochirus cavimanus*	83.448	943
sturgeon poacher	*Podothecus accipenserinus*	78.832	744
N/A	*Strongylocentrotus sp.*	74.491	566
Alaska skate	*Arctoraja parmifera*	57.58	9
Pacific halibut	*Hippoglossus stenolepis*	56.074	25
starry flounder	*Platichthys stellatus*	54.583	57
armorhead sculpin	*Gymnocanthus galeatus*	41.67	123
blackspined sea star	*Lethasterias nanimensis*	36.868	100
flathead sole	*Hippoglossoides elassodon*	21.839	47
plain sculpin	*Myoxocephalus jaok*	15.32	19
fuzzy hermit crab	*Pagurus trigonocheirus*	15.087	377
Tanner crab	*Chionoecetes bairdi*	13.016	45
Bering beringius	*Beringius behringi*	12.258	94
N/A	*Plicifusus kroyeri*	11.315	189
big skate	*Beringraja binoculata*	9.62	1
great sculpin	*Myoxocephalus polyacanthocephalus*	9.6	6
Alaska plaice	*Pleuronectes quadrituberculatus*	9.573	10
giant sea star	*Evasterias echinosoma*	9.328	6
basketstar	*Gorgonocephalus eucnemis*	8.362	40
Fisher sun star	*Solaster sp. F (Clark)*	7.38	9
graceful decorator crab	*Oregonia gracilis*	7.148	211
Pacific capelin	*Mallotus catervarius (=villosus)*	7.02	386
Alaskan hermit	*Pagurus ochotensis*	6.6	94
roughspine sculpin	*Triglops macellus*	4.756	80
searcher	*Bathymaster signatus*	4.649	21

Common Name	Scientific Name	Weight (kg)	Number
Kamchatka flounder	*Atheresthes evermanni*	4.587	43
N/A	*Chrysaora melanaster*	3.237	18
mottled anemone	*Urticina crassicornis*	2.962	4
sea raspberry	*Gersemia sp.*	2.918	N/A
threaded sculpin	*Gymnocanthus pistilliger*	2.874	79
Bering skate	*Bathyraja interrupta*	2.74	1
N/A	*Pteraster sp.*	1.396	6
compound ascidian unid.	*N/A*	0.943	N/A
Arctic surfclam	*Mactromeris polynyma*	0.636	4
sawback poacher	*Sarritor frenatus*	0.384	6
Pacific lyre crab	*Hyas lyratus*	0.351	13
parma sand dollar	*Echinarachnius parma*	0.351	9
Alaska skate egg case	*Arctoraja parmifera egg case*	0.322	6
snow crab	*Chionoecetes opilio*	0.264	1
Alaska great-tellin	*Tellina lutea*	0.197	4
whitespotted greenling	*Hexagrammos stelleri*	0.154	4
Pacific spiny lumpsucker	*Eumicrotremus orbis*	0.107	3
ribbed sculpin	*Triglops pingelii*	0.066	4
sand lance unid.	*Ammodytes sp.*	0.022	9
N/A	*Crangon sp.*	0.009	4
empty bivalve shells	*N/A*	0.009	N/A

Table 2: EBS Voucher and Otolith Collection Specimens: Within 3-mile Territorial Waters

Common Name	Scientific Name	count	category
No Vouchers Collected	*N/A*	0	VOUCHER
arrowtooth flounder	*Atheresthes stomias*	3	OTOLITH
Kamchatka flounder	*Atheresthes evermanni*	9	OTOLITH
flathead sole	*Hippoglossoides elassodon*	3	OTOLITH
yellowfin sole	*Limanda aspera*	3	OTOLITH
northern rock sole	*Lepidopsetta polyxystra*	8	OTOLITH
Alaska plaice	*Pleuronectes quadrituberculatus*	2	OTOLITH
Pacific cod	*Gadus macrocephalus*	8	OTOLITH
walleye pollock	*Gadus chalcogrammus*	3	OTOLITH

Citations:

DeFilippo, L., Kotwicki, S., Barnett, L., Richar, J., Litzow, M. A., Stockhausen, W. T., & Palof, K. (2023). Evaluating the impacts of reduced sampling density in a systematic fisheries-independent survey design. *Frontiers in Marine Science*, 10, 1219283.

Markowitz, E. H., Rohan, S. K., Wassermann, S., Charriere, N. E., Anderson, C. B., and Stevenson, D. E. (2025). Results of the 2024 eastern Bering Sea continental shelf bottom trawl survey of groundfish and invertebrate fauna. U.S. Department of Commerce, NOAA Technical Memorandum NMFS-AFSC-499, 203 p.